

IT Project Management

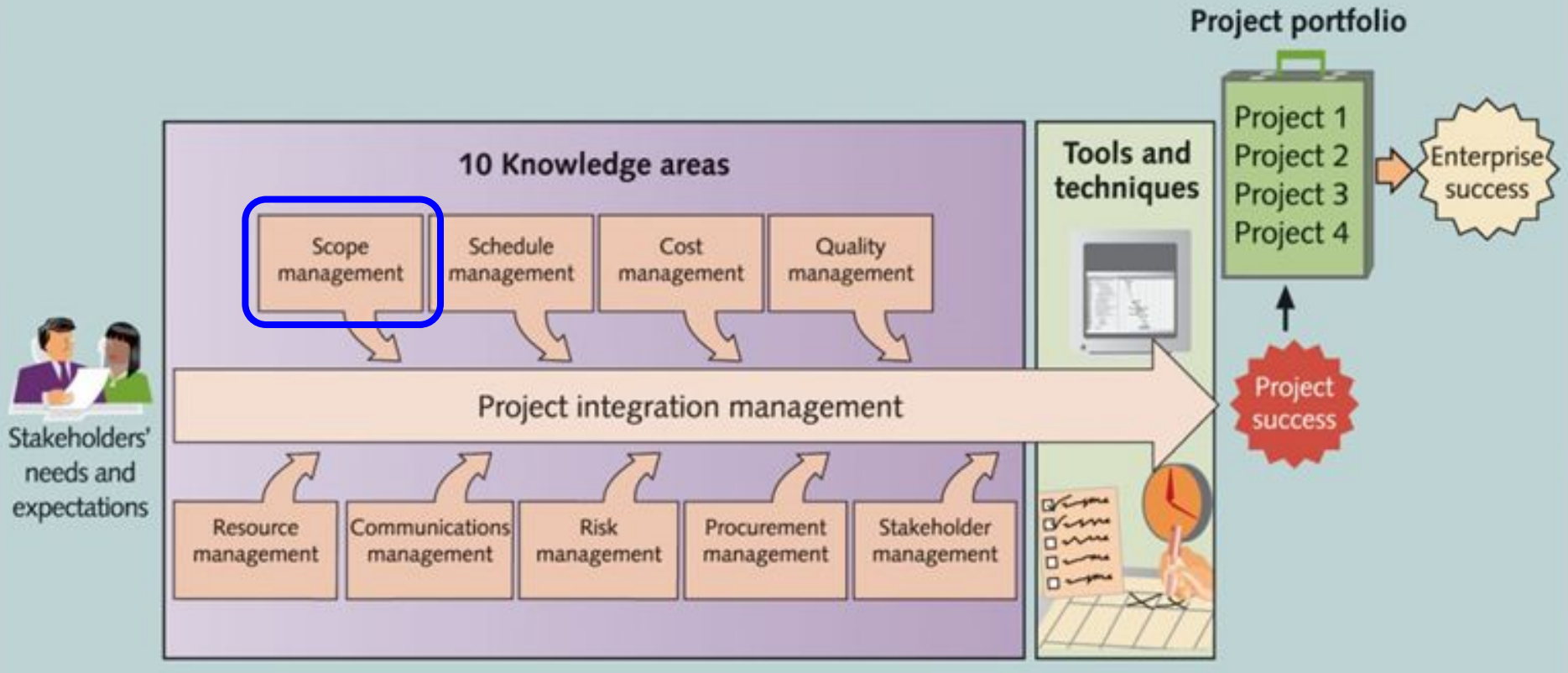
Lecture 4: Project **Scope** Management

—Defining & Controlling Work Boundaries—

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Project Scope Management



About this lecture



Learning objectives:

- Understand **what project scope management is**
- Learn the **processes of scope management** in a project
- Explore **project selection methods** and scope planning
- Understand **Work Breakdown Structure (WBS)** and scope definition

What is Scope Management?

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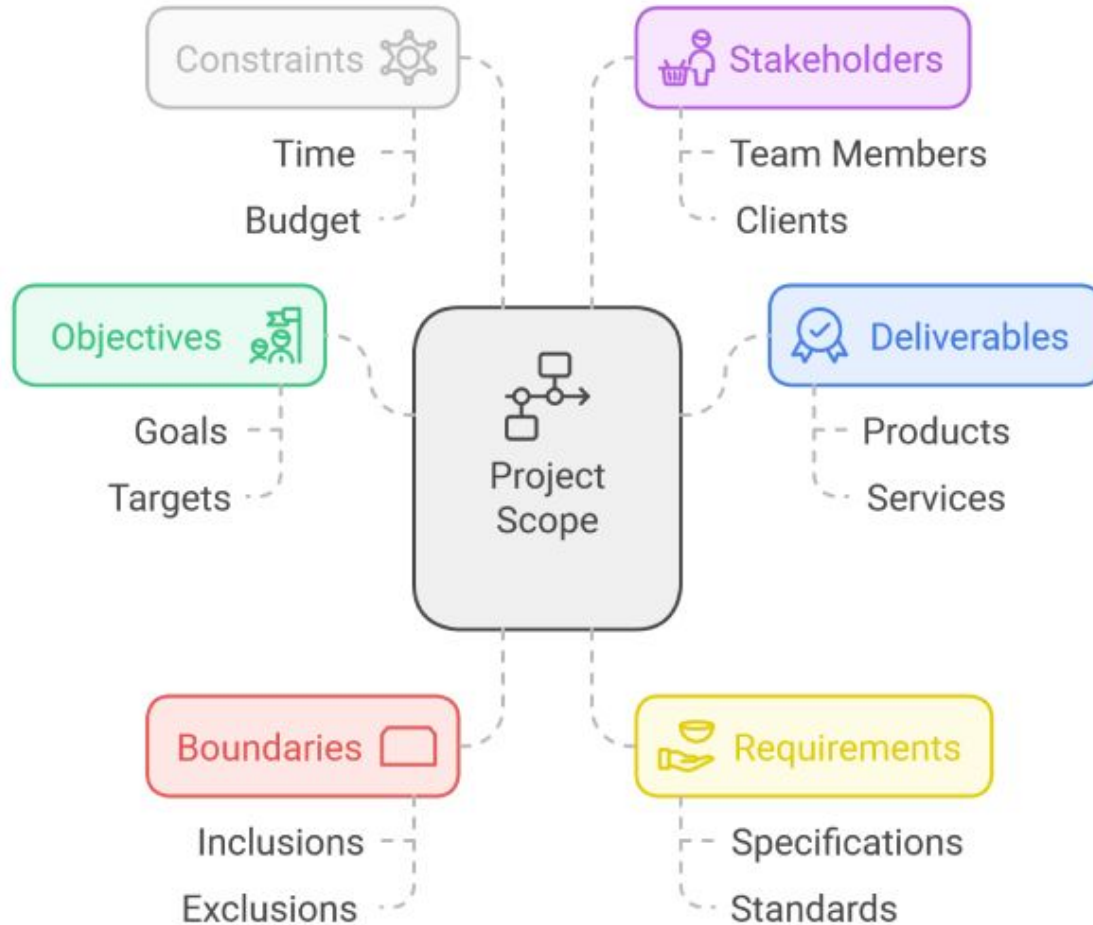
- **Scope:** all work required to **deliver the project's product/service**

It defines **what will be done and not be done** in the project

Includes:

- **Intermediate and final deliverables** (→ real outputs)
 - **Constraints** (→ realistic planning)
 - **Assumptions** (→ risk identification)
 - **Requirements** (→ shareholders' expectations)
- **Scope management:** ensure all **stakeholders agree on the scope**

Key Components of Scope Management



The 4 Key Steps of Project Scope Management

1. **Initiation:** Select project
 - Financial analysis
 - Project Charter
2. **Scope Planning:** Define & document project scope
 - Scope Statement
3. **Scope Definition:** Break down work into tasks
 - Work Breakdown Structure (WBS)
4. **Scope Verification & Control:** Ensure deliverables meet requirements

1. Initiation

1. Initiation: Project Selection

Steps to select a Project:

1. Align with **business strategy** (→ strategic level)
2. Identify **business needs** & potential projects (→ tactical level)
3. Evaluate **resources, budget, and feasibility** (→ constraints, etc.)
4. Select the most suitable project using **selection methods**

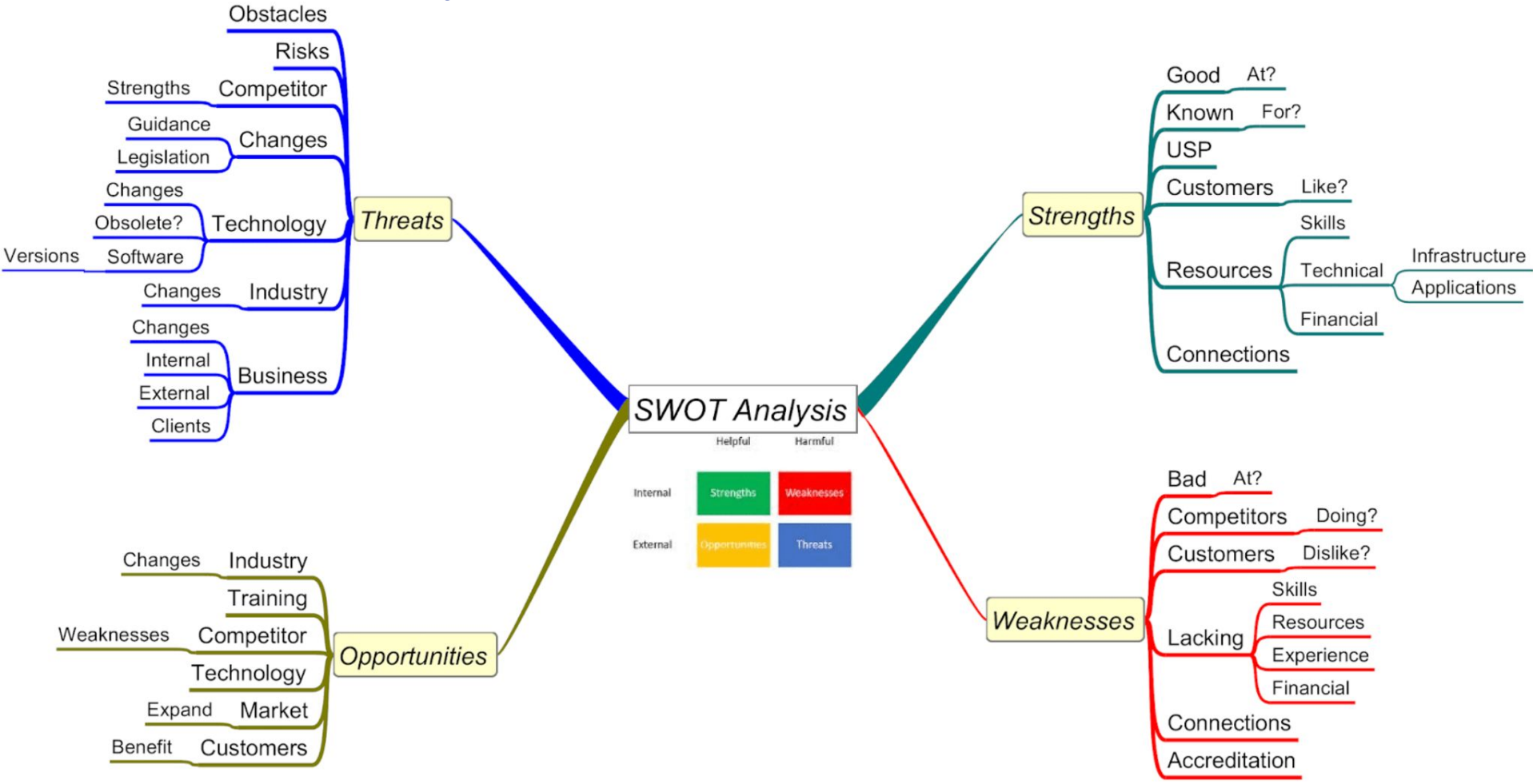
1. Initiation: Project Selection

SWOT Analysis

A tool for strategic analysis and planning:

- **Strengths** – Internal advantages that create competitive benefits
- **Weaknesses** – Internal limitations that may hinder success
- **Opportunities** – External factors that can drive growth
- **Threats** – External risks that may impact the organization

1. Initiation: Project Selection



1. Initiation: Project Selection

Common Project **Selection Methods**

Financial analysis methods:

- Net Present Value (NPV) Analysis
- Return on Investment (ROI)
- Payback Analysis

Other strategic methods:

- Weighted Scoring Model (WSM)
- Balanced Scorecard

1. Initiation: Project Selection

Method	Definition	Key Benefit	Key Limitation	Best Used For
Net Present Value (NPV)	Calculates the present value of future cash flows to determine project profitability.	Accounts for time value of money .	Requires an accurate discount rate estimation.	Long-term, high-investment projects.
Return on Investment (ROI)	Measures the profitability of a project relative to its cost.	Simple way to quickly compare different projects.	Oversimplified , lacks timeframe information.	Comparing multiple investment options quickly.
Payback Analysis	Calculates the time required to recover the initial investment.	Easy to understand and useful for short-term decision-making .	Ignores cash flows after the payback period.	Projects requiring fast cost recovery.
Weighted Scoring Model (WSM)	Assigns weights to different project criteria (cost, risk, strategic alignment, etc.), then scores projects accordingly.	Helps in multi-criteria decision-making .	Subjective ; depends on how weights are assigned.	Evaluating projects with multiple competing factors.
Balanced Scorecard	Evaluates projects based on financial, customer, internal process, and innovation perspectives .	Ensures a holistic project evaluation beyond just financial metrics.	More complex to implement than other methods.	Strategic decision-making and alignment with business goals.

1. Initiation: Project Selection

NPV: Example

	A	B	C	D	E	F	G
1	Discount rate	10%					
2							
3	PROJECT 1	YEAR 1	YEAR 2	YEAR 3	YEAR 4	YEAR 5	TOTAL
4	Benefits	\$0	\$2,000	\$3,000	\$4,000	\$5,000	\$14,000
5	Costs	\$5,000	\$1,000	\$1,000	\$1,000	\$1,000	\$9,000
6	Cash flow	(\$5,000)	\$1,000	\$2,000	\$3,000	\$4,000	\$5,000
7	NPV	\$2,316					
8		Formula =npv(b1,b6:f6)					
9							
10	PROJECT 2	YEAR 1	YEAR 2	YEAR 3	YEAR 4	YEAR 5	TOTAL
11	Benefits	\$1,000	\$2,000	\$4,000	\$4,000	\$4,000	\$15,000
12	Costs	\$2,000	\$2,000	\$2,000	\$2,000	\$2,000	\$10,000
13	Cash flow	(\$1,000)	\$0	\$2,000	\$2,000	\$2,000	\$5,000
14	NPV	\$3,201					
15		Formula =npv(b1,b13:f13)					
16							
17	RECOMMEND PROJECT 2 BECAUSE IT HAS THE HIGHER NPV.						

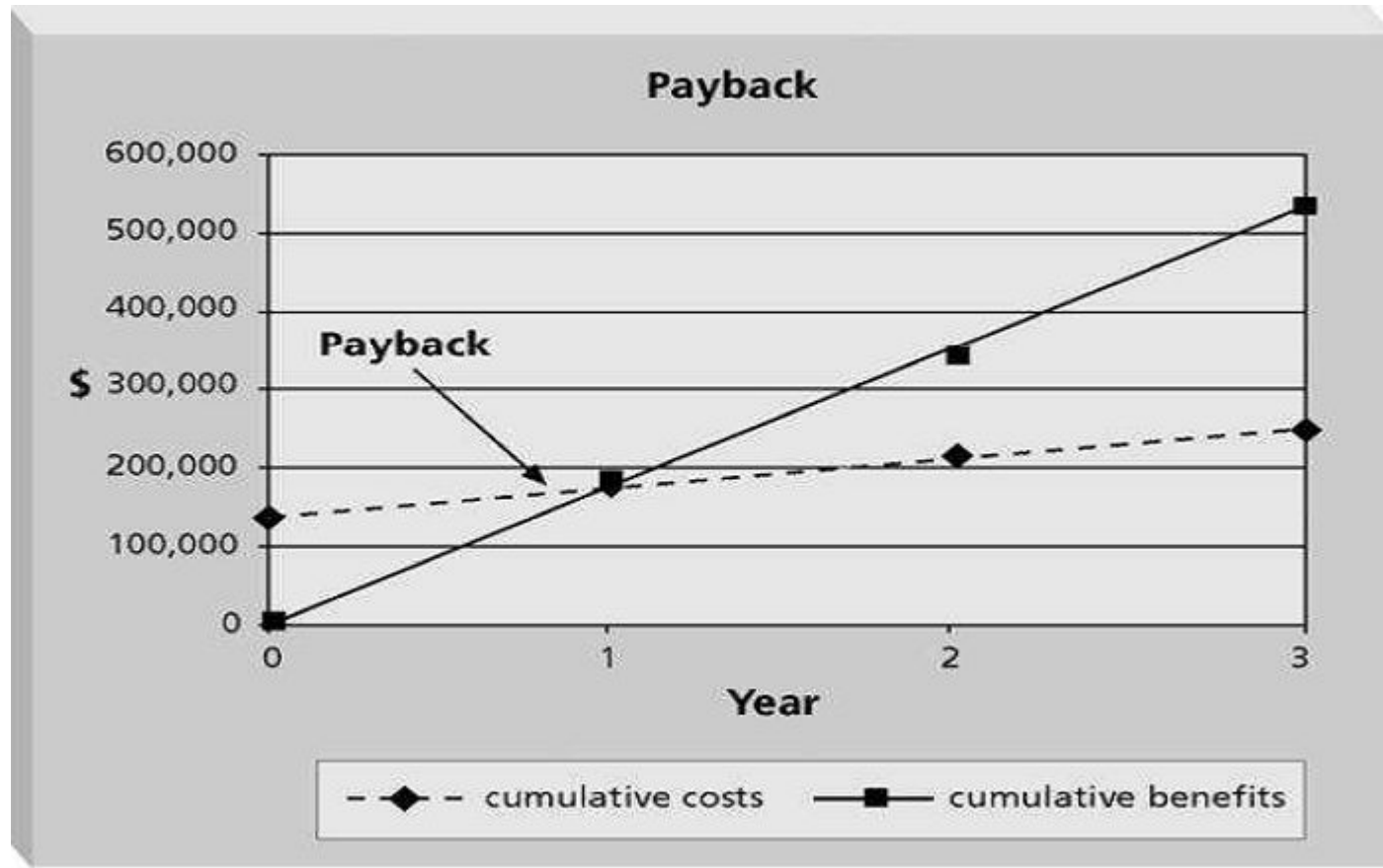
1. Initiation: Project Selection

ROI: Example

Discount rate	8%						
Assume the project is completed in Year 0			Year				
	0	1	2	3	Total		
Costs	140,000	40,000	40,000	40,000			
Discount factor	1	0.93	0.86	0.79			
Discounted costs	140,000	37,200	34,400	31,600	243,200		
Benefits	0	200,000	200,000	200,000			
Discount factor	1	0.93	0.86	0.79			
Discounted benefits	0	186,000	172,000	158,000	516,000		
Discounted benefits - costs	(140,000)	148,800	137,600	126,400	272,800	← NPV	
Cumulative benefits - costs	(140,000)	8,800	146,400	272,800			
ROI	112%						
	Payback before Year 1						

1. Initiation: Project Selection

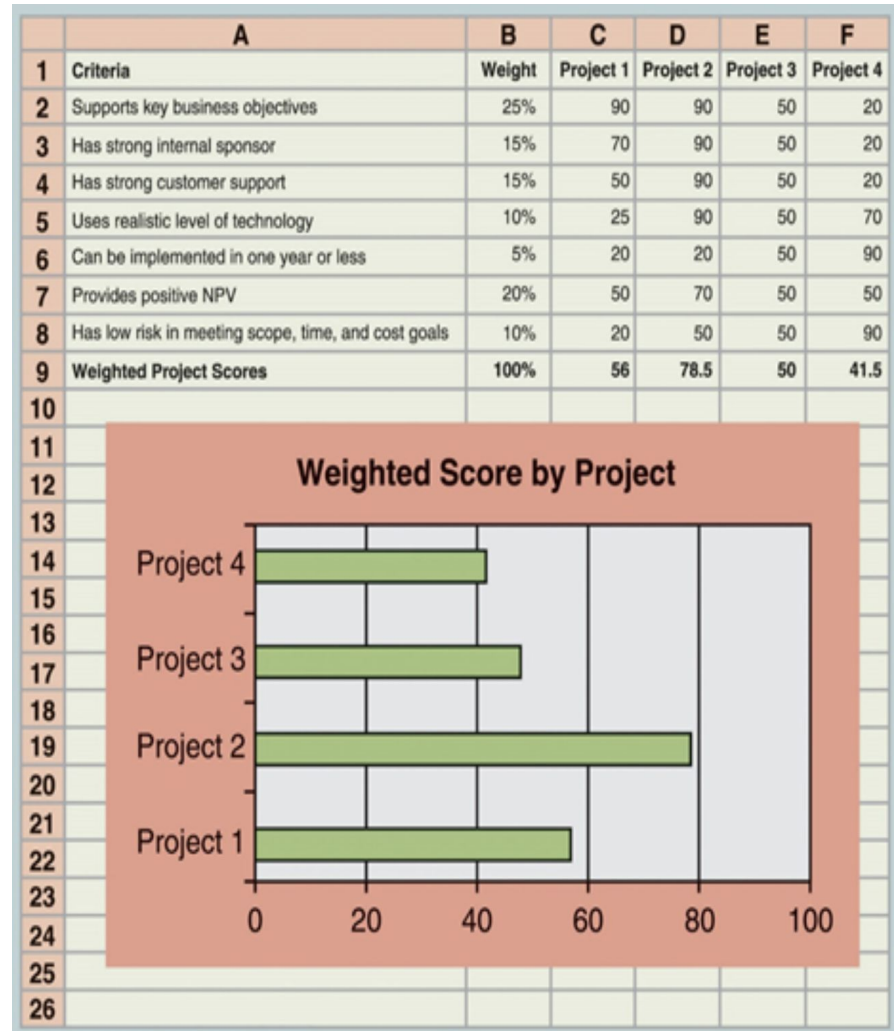
Payback Analysis: Example



1. Initiation: Project Selection

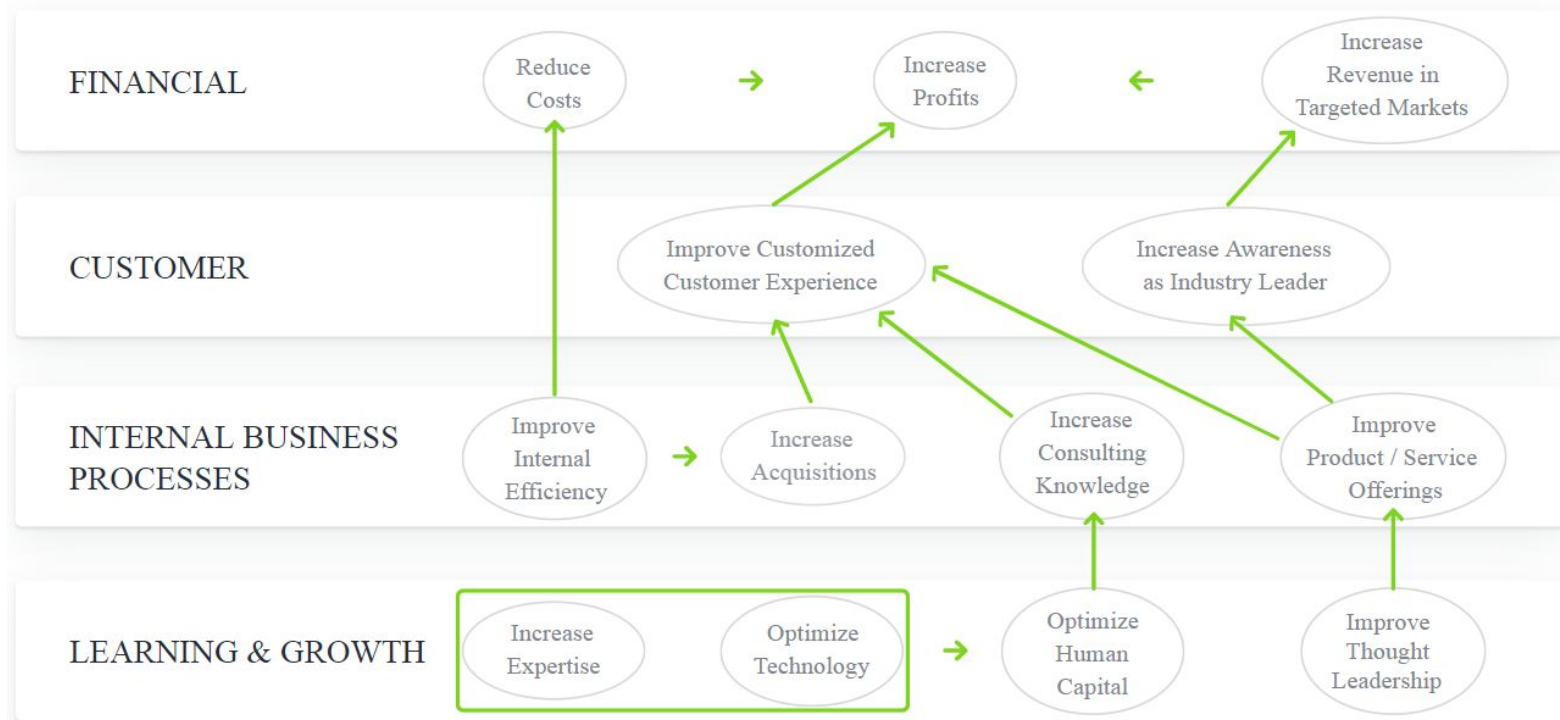
Weighted Scoring Model (WSM):

Example



1. Initiation: Project Selection

Balanced Scorecard: Example



1. Initiation: Project Selection

Using the Right Selection Method:

- **NPV** → If the project has **long-term financial impacts**
- **ROI** → If you need a **quick profitability comparison**
- **Payback Analysis** → If you want to know **how fast you'll recover costs**
- **WSM** → If the decision involves **multiple factors beyond finance**
- **Balanced Scorecard** → If the project must **align with broader business goals**

1. Initiation: Project Charter

What is a Project Charter?

- A formal document that **authorizes a project's existence**
- Signed by **key stakeholders** to confirm project approval
- Defines project **objectives, scope, stakeholders, constraints, etc.**

1. Initiation: Project Charter

Key Elements of a Project Charter:

- Project objectives
- Major deliverables
- Project timeline
- Roles and responsibilities
- etc.

1. Initiation: Project Charter

ONE-PAGE PROJECT CHARTER TEMPLATE

PROJECT NAME		PROJECT MANAGER	PROJECT SPONSOR
EMAIL	PHONE	ORGANIZATIONAL UNIT	
	000-000-0000		
ESTIMATED COSTS	EXPECTED SAVINGS	EXPECTED START DATE	EXPECTED COMPLETION
\$0	\$0	00/00/0000	00/00/0000

PROJECT OVERVIEW

PROBLEM OR ISSUE	
PURPOSE OF PROJECT	
BUSINESS CASE	
GOALS / METRICS	
EXPECTED DELIVERABLES	

PROJECT SCOPE

WITHIN SCOPE	
OUTSIDE OF SCOPE	

TENTATIVE SCHEDULE

KEY MILESTONE	START	FINISH
Form Project Team / Preliminary Review / Scope	00/00/0000	00/00/0000
Finalize Project Plan / Charter / Kick Off	00/00/0000	00/00/0000
Define Phase	00/00/0000	00/00/0000
Measurement Phase	00/00/0000	00/00/0000
Analysis Phase	00/00/0000	00/00/0000
Improvement Phase	00/00/0000	00/00/0000
Control Phase	00/00/0000	00/00/0000
Project Summary Report and Close Out	00/00/0000	00/00/0000

Project Charter Example

Project Name	IVR Project		
Project Sponsor	Dave Sponsor	Project Manager	Alice Michaels
Date of Project Approval	8th Mar 2015	Last Revision Date	17th Apr 2015

Project Description	To introduce a new automated telephone system to ensure all calls get answered.		
Scope	A IVR system will be introduced to assist the sales team in taking orders, and also to ensure no orders are missed. The system is only to help the sales team at this stage, other teams such as support are out of scope.		
Business Case	To increase orders per sales team member by 20% from current levels. To reduced unhandled calls to 0%. To increase customer satisfaction by 10 points.		
Constraints (in priority order)	Time	4 months	
	Budget	4 developers + 1 sales team rep	
	Scope	TBD	
	Quality	Prioritize time & budget over quality	
Project Deliverables	An IVS system to assist the sales team + training for the sales team + support during the first operational month of the system.		
Benefits (measurable results)	See KPIs below + business case above		
	KPI	Baseline	Goal
	Orders per sales person pd	20	24
	Unhandled calls pd	11	0
	Customer satisfaction	17	27
Steering Committee	CEO	Project Team	Sales Rep
	Finance Director		4x developers TBD
	Sales Director		
Key Stateholders	Name	Success Criteria	
Risks	No team members have any previous experience of IVR setup, so there is a chance we've hugely underestimated the work involved.		

2. Scope Planning

2. Scope Planning

- **Scope Planning** → Creating documents (scope statement, etc.)
that **define and control scope**
- **Scope Statement Includes:**
 - Project justification (why the project exists)
 - High-level description of the product/service
 - List of deliverables
 - Success criteria
 - etc.

2. Scope Planning: Scope Statement

Project Scope Statement			
Title	Timesheet Reporting Tool	Date	June 08, 2017
Project Manager	Nicole Hansen		
Project Justification	The Timesheet Reporting tool will be the creation of a user-friendly web-based application. The application will automate the process of tracking the daily time entries made by employees		
Project Scope Description	The timesheet reporting tool will be used as a tracking tool to review the time entries by the Information Technology employees globally		
Project Objective	To create a tool to track the timesheet entries for all the Information Technology employees		
High Level Requirements	- User access for the tool granted to all employees - Admin access to support team - Report generation access to project managers - Enable users to remotely access the tool - Standard templates for all reports		
In Scope	- Multi – level approvals for timesheets - Timesheet tasks additions to the tool by Admin access users 'Copy Previous Week' timesheets feature - Access tool via Desktop		

Project Scope Statement Example			
Project Name	IVR Project		
Project Sponsor	Dave Sponsor	Project Manager	Alice Michaels
Date of Project Approval	08 March 2015	Last Revision Date	08 March 2015
Scope Description	IN SCOPE: - An IVR system to handle and direct sales calls. - Setup of the IVR system ready to handle all sales enquiries - Training for the sales team on how to use the system - An administration system so the IVR system can be configured by the sales team OUT OF SCOPE: - Support for any other team other than sales - The ability to route complex sales enquiries, such as partnerships etc.		
Project Deliverables	- A customer facing IVR system - An IVR configuration system - A training manual for new and existing sales team members		
Acceptance Criteria	The project will be accepted when it is agreed by the sale team that the deliverables meet their needs (deliverables signed off). The project will also be considered a success if post launch testing shows that customer satisfaction is over 5 points higher than where it is now.		
Constraints	The sales team are not dedicated to this project, and must fit this around their day-to-day sales jobs.		
Assumptions	A sales team & full project team will be available to co-create the WBS		

3. Scope Definition

3. Scope Definition

- Define work in detail by **breaking it into smaller manageable tasks**
- **Benefits:**
 - Improves **accuracy** in time, cost, and resource **estimation**
 - Provides a **basis for performance measurement**
 - Clearly communicates **responsibilities**

3. Scope Definition: Work Breakdown Structure (WBS)

What is a Work Breakdown Structure (WBS)?

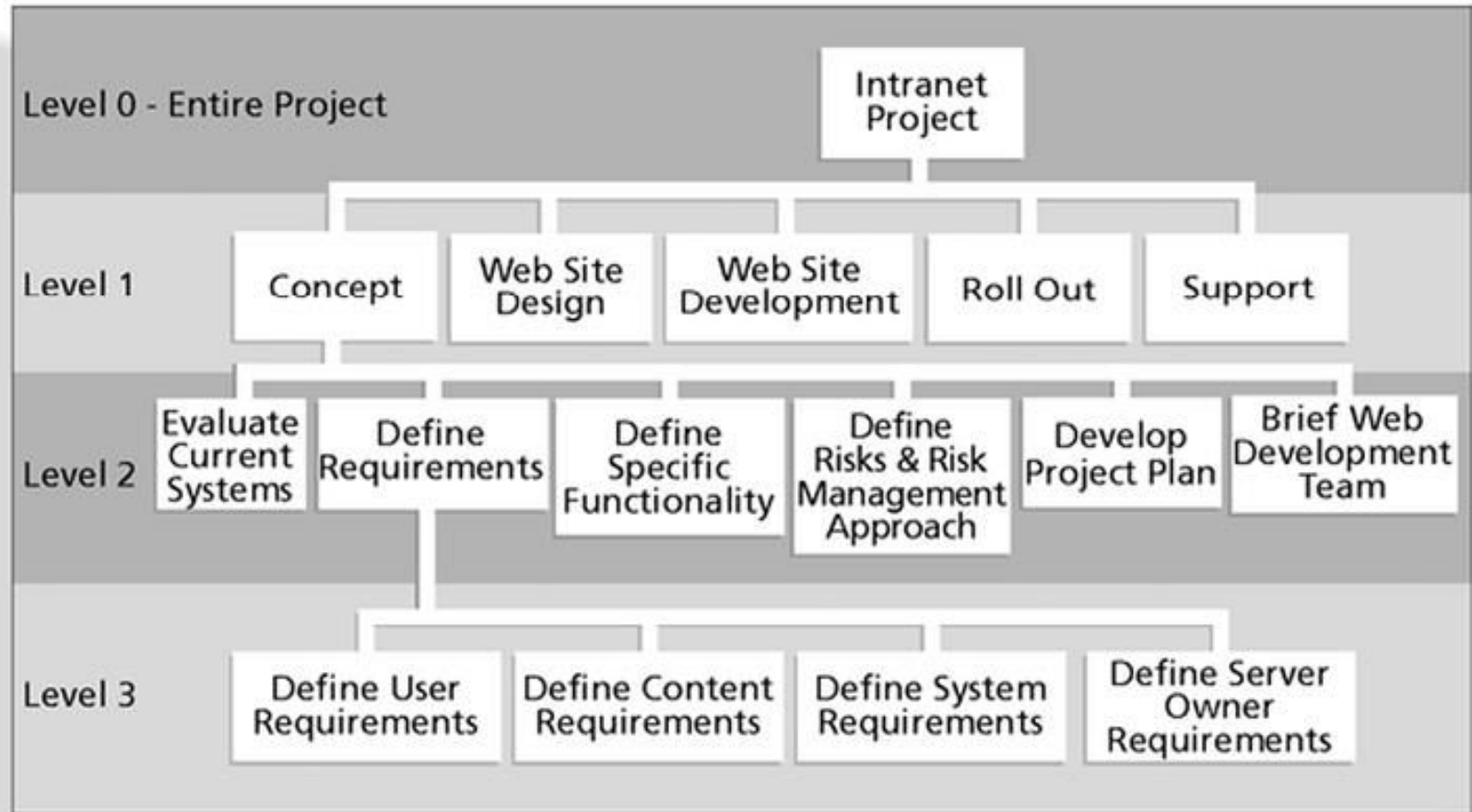
- A **hierarchical breakdown** of project work into **smaller tasks**
- Helps in **estimating costs, schedules, and assigning responsibilities**

3. Scope Definition: Work Breakdown Structure (WBS)

Key elements to consider when creating WBS:

- **List of Deliverables** – Organized hierarchically
- **Time Estimations** – Estimated for each task
- **Task Scheduling** – Start and end dates for each task
- **Resource Allocation** – Assigning human and material resources
- **Cost Estimation** – Budgeting for each phase
- **Risk Management** – Identifying potential risks

3. Scope Definition: Work Breakdown Structure (WBS)



3. Scope Definition: Work Breakdown Structure (WBS)

1.0 Concept

1.1 Evaluate current systems

1.2 Define Requirements

1.2.1 Define user requirements

1.2.2 Define content requirements

1.2.3 Define system requirements

1.2.4 Define server owner requirements

1.3 Define specific functionality

1.4 Define risks and risk management approach

1.5 Develop project plan

1.6 Brief Web development team

2.0 Web Site Design

3.0 Web Site Development

4.0 Roll Out

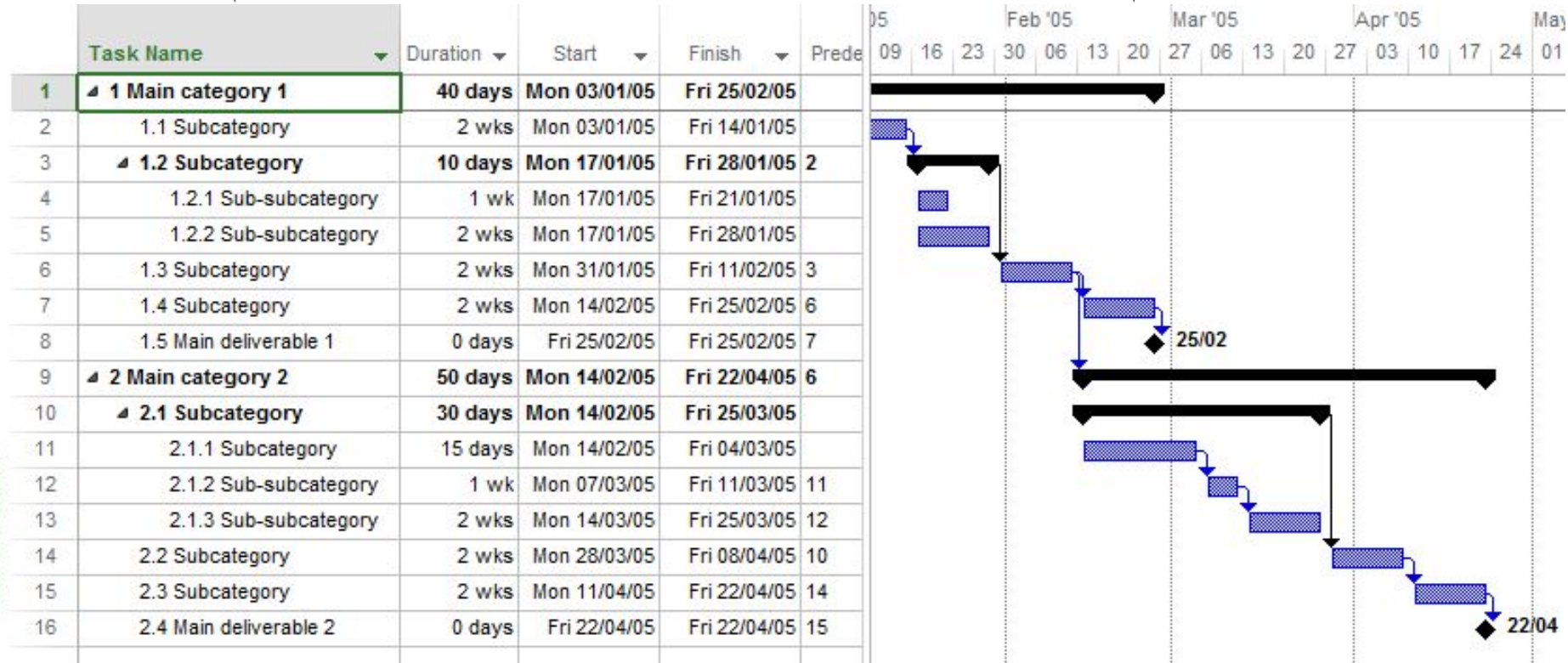
5.0 Support

3. Scope Definition: Work Breakdown Structure (WBS)

WBS

Schedule (Gantt chart)

GANTT CHART



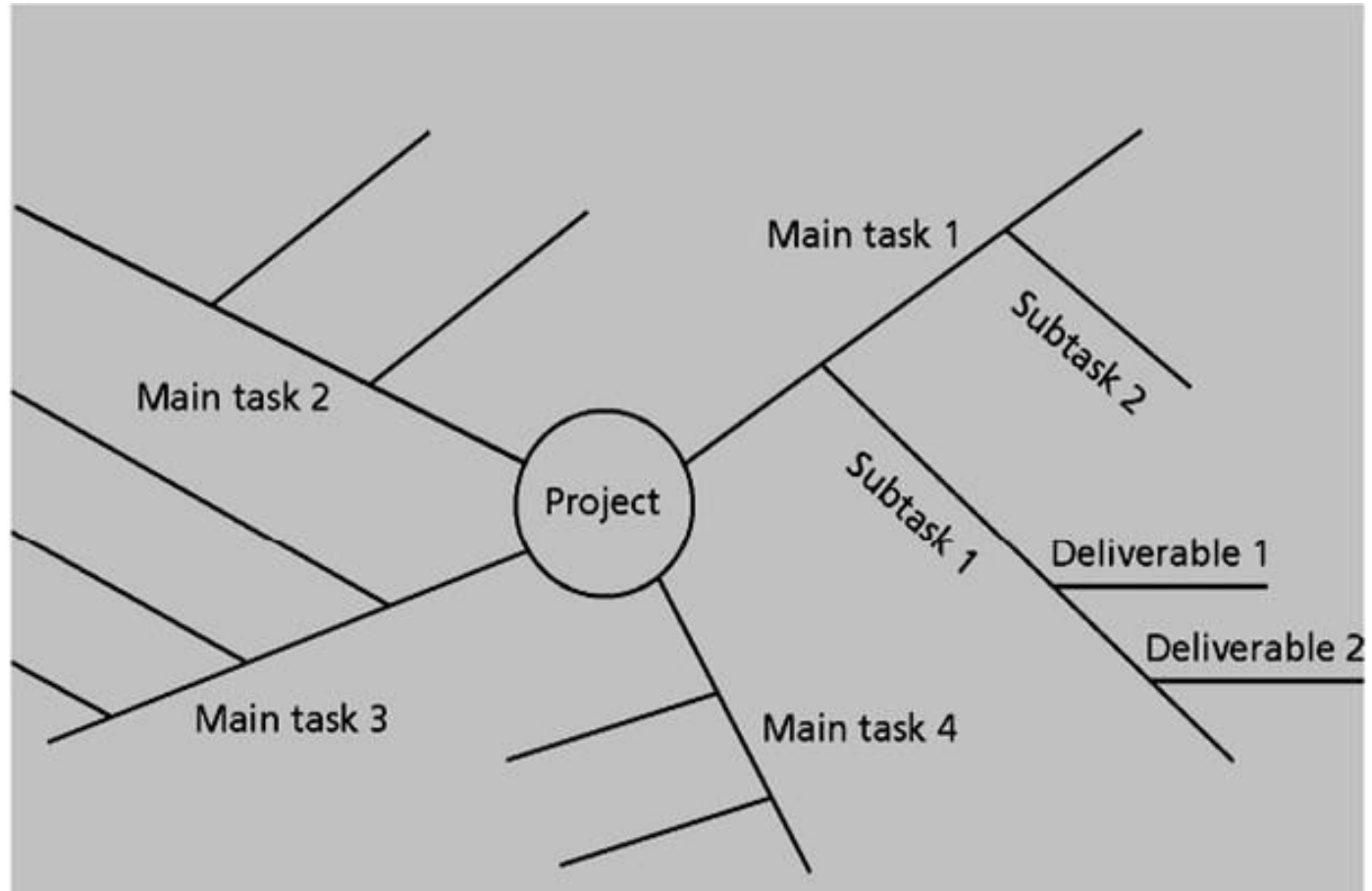
3. Scope Definition: Work Breakdown Structure (WBS)

Approaches to WBS Development:

- **Top-Down Approach** → Start with the big picture, then break it down
- **Bottom-Up Approach** → Start with small tasks, group them into larger units
- **Mind-Mapping Approach** → Visualizing tasks in a non-linear structure
- **Analogous Approach** → Reusing WBS from similar projects

3. Scope Definition: Work Breakdown Structure (WBS)

Example:
Mind-Mapping
Approach



3. Scope Definition: Work Breakdown Structure (WBS)

Principles of WBS:

1. Each task should only appear **once** in the WBS
2. The **sum of sub-tasks** must equal the parent task
3. Each task should be **independent, measurable, and clearly defined**
4. WBS should be **consistent with the development process (e.g., agile iters)**
5. WBS should be **developed with input from project team members**
6. **Documentation** should be included for each WBS item for clarity
7. WBS should be **flexible for changes** when needed

3. Scope Definition: Work Breakdown Structure (WBS)

WBS Verification Checklist:

- Is every task **measurable**?
- Are tasks **independent**?
- Can time, resource, and cost be **estimated**?
- Is the completion **timeline within limits**?

→ If any “No” answer: Further Breakdown Required

4. Scope Verification & Control

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 **Scope Creep:** Uncontrolled **changes**

that **expand project scope** beyond the original plan

 **Goal:** To prevent **Scope Creep**

4. Scope Verification & Control

Common Causes of Scope Creep:

- Unclear requirements
- Poor communication
- Lack of processes to control change

4. Scope Verification & Control

How to Prevent Scope Creep?

- Clearly define and document scope **before starting work**
- Use a **Change Control Process** for approving scope changes
- Regular **scope verification** to keep work on track

Financial Analysis for Project Selection

Financial Analysis for Project Selection

Financial Analysis Methods for Evaluation:

- **Net Present Value (NPV)** → Determines project profitability
- **ROI (Return on Investment)** → Measures return on project cost
- **Payback Analysis** → Calculates how long to recover costs

Financial Analysis for Project Selection

Key Financial Concepts:

- **Time Value of Money** → Money today is worth more than in the future

- **Future Value Formula:** $V_n = V_0(1 + r)^n$

Where:

V_n : value at year n

V_0 : value at year 0

r : interest rate ($r \geq 0$)

n : project life expectancy

- **Present Value Formula:** $V_0 = V_n / (1 + r)^n$

Financial Analysis for Project Selection

Key Financial Concepts:

$$NPV = \sum_{t=1}^n \frac{B_t - C_t}{(1 + r)^t}$$

Net, Present, Value

Net Value at year t

Net present value of year t

Sum all the years

Compound interest rate at year t

Where:

- B_t: benefit in year t
- C_t: cost in year t
- r: interest rate ($r \geq 0$)
- n: project life expectancy

Financial Analysis for Project Selection

Key Financial Concepts:

Notice:
Net (Profit, Value)
vs. Benefit, Profit, Value

Net Profit = Net Value = Benefit - Cost
(in **present value's** term)

$$\text{ROI} = \frac{\text{NetProfit}_{\text{Present}}}{\text{InvestmentCost}_{\text{Present}}} * 100\%$$

Return on Investment

Investment Cost
(in **present value's** term)

Financial Analysis for Project Selection

NPV: Example

	A	B	C	D	E	F	G
1	Discount rate	10%					
2							
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7	NPV	\$2,316					
8		Formula =npv(b1,b6:f6)					
9							
10	PROJECT 2	YEAR 1	YEAR 2	YEAR 3	YEAR 4	YEAR 5	TOTAL
11	Benefits	\$1,000	\$2,000	\$4,000	\$4,000	\$4,000	\$15,000
12	Costs	\$2,000	\$2,000	\$2,000	\$2,000	\$2,000	\$10,000
13	Cash flow	(\$1,000)	\$0	\$2,000	\$2,000	\$2,000	\$5,000
14	NPV	\$3,201					
15		Formula =npv(b1,b13:f13)					
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17	RECOMMEND PROJECT 2 BECAUSE IT HAS THE HIGHER NPV.						

Financial Analysis for Project Selection

ROI: Example

Discount rate	8%					
Assume the project is completed in Year 0			Year			
	0	1	2	3	Total	
Costs	140,000	40,000	40,000	40,000		
Discount factor	1	0.93	0.86	0.79		
Discounted costs	140,000	37,200	34,400	31,600	243,200	
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Discounted benefits - costs	(140,000)	148,800	137,600	126,400	272,800	← NPV
Cumulative benefits - costs	(140,000)	8,800	146,400	272,800		
ROI	112%					
Payback before Year 1						

Exercise

Financial Analysis for Project Selection

Exercise

Exercise 1

Mr. M has to pay Mr. N 1,000 USD in the second year and 3,000 USD in the fifth year from the present time. If they renegotiate the contract to pay off the entire amount in the third year, how much does Mr. M need to pay? Assume an interest rate of 6% per year.

Exercise 2

A computer is sold for 2,000 USD with an additional 12-month installment plan of 250 USD per month, based on an 18% monthly interest rate. How much would it cost to buy the computer if paid in full immediately?

Financial Analysis for Project Selection

Exercise

Exercise 3*

Mr. A has made seven annual payments of 10 million VND at the end of each year for a loan of 100 million VND, with an interest rate of 5% per year. If he wants to pay off the remaining amount over the next five years, how much must he pay annually?

Hint: New loan principal = Previous principal + Interest - Payment
(expand it and derive closed-form solution, see appendix for formula)

Exercise 4

You rent out a house for 6,000 USD per year, with payments made at the end of each year for five years. The entire rental income is deposited in a bank with a compound interest rate of 6% per year. After five years, what will be the total amount, including principal and interest?

Financial Analysis for Project Selection

Exercise

Exercise 5

We are considering two projects, A and B, with 1,000 USD as initial investment cost in the first year (Year 0) for each project, and no other cost for other years (Year 1 to Year 4). The estimated benefits for each project are as follows:

Project	Year 1	Year 2	Year 3	Year 4
A	500	500	500	500
B	800	800	200	200

Calculate the **Net Present Value (NPV)** and **Return on Investment (ROI)** to determine which project should be selected. Assume an interest rate of **10%**.

Financial Analysis for Project Selection

Exercise

Exercise 6

Two projects, A and B, have been estimated with the following data:

Project A	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
Investment	500	0	100	0	0	0
Benefit	0	250	350	450	500	-100

Project B	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
Investment	100	0	50	0	0	0
Benefit	0	-100	200	200	200	300

Given an interest rate of **10%**, calculate the **NPV** and **ROI** for each project.

Summary



You have learned to:

- Understand **what project scope management is**
- Learn the **processes of scope management** in a project
- Explore **project selection methods** and scope planning
- Understand **Work Breakdown Structure (WBS)** and scope definition

Thank you for listening

Q & A

Appendix

Financial Analysis for Project Selection

Amortization formula: To calculate the annual payment amount for a loan

We can inverse this formula
to calculate P from A



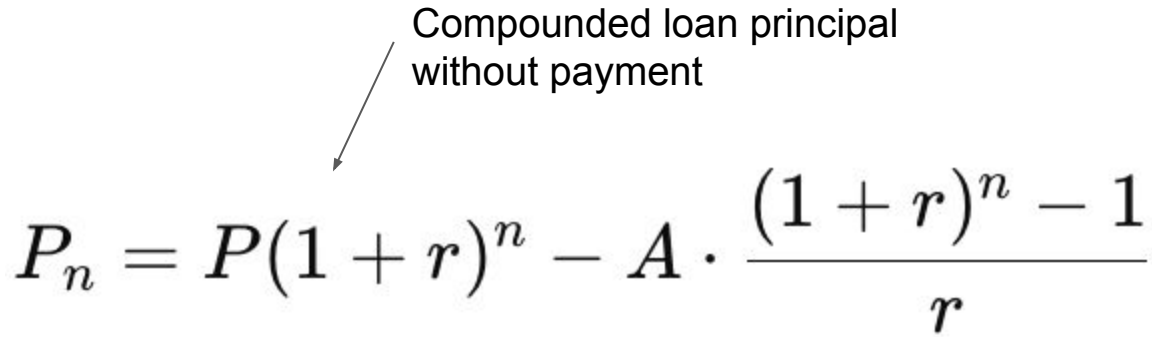
$$A = \frac{Pr}{1 - (1 + r)^{-n}}$$

Where:

- A = Annual payment
- P = Initial loan principal
- r = Annual interest rate
- n = Number of years

Financial Analysis for Project Selection

To calculate the remaining loan principal after n years


$$P_n = P(1 + r)^n - A \cdot \frac{(1 + r)^n - 1}{r}$$

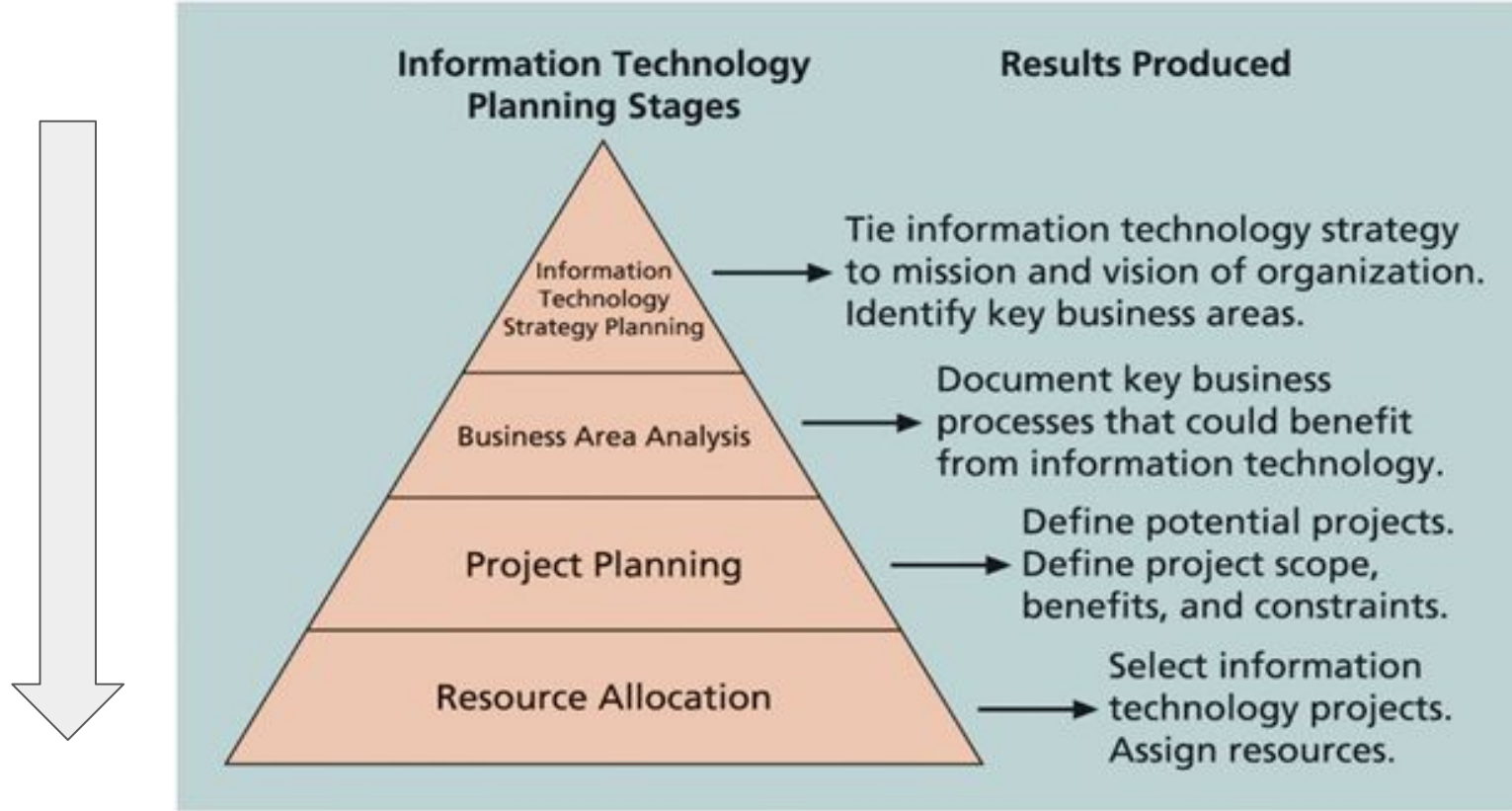
Compounded loan principal
without payment

Amount paid after n years
(related to amortization formula, but
calculate future values of P and A)

Where:

- P_n = Remaining loan principal after n years
- P = Initial loan principal
- r = Annual interest rate
- A = Annual payment
- n = Number of years

1. Initiation: Project Selection (PMI Guide)



Planning process for selecting IT projects